

School of AI Portfolio

Generated 2026-05-18 08:45. 5 projects ranked by complexity and documentation depth. Featured: knowledge-graph, task-link, stock-intel.

1. System Design Knowledge Graph

Assignment 5 | Full-stack AI | Complexity 100 | Info 100

<https://github.com/varunsood189/system-design-knowledge-graph>

FastAPI | NetworkX | trafilatura | FastMCP | Gemini JSON planner | optional LLM Gateway V2

An AI-powered evolving knowledge graph for system design learning. Ingest engineering blog URLs, extract concepts and relationships with LLM agents, persist an evolving graph in data/graph.json, and generate contextual learning recommendations.

Tech: FastAPI, Gemini, MCP, FastMCP, NetworkX, Pydantic, LLM, trafilatura, Gemini JSON planner, optional LLM Gateway V2

- LLM orchestrator: Gemini chooses MCP tools each turn (extract ? concepts ? relationships ? merge ? recommendations)
- Live agent log in the UI via Server-Sent Events (POST /ingest/stream)
- Structured JSON agents: Pydantic-validated concept, relationship, and recommendation outputs (prompts/*.txt)
- Persistent graph: NetworkX merged into data/graph.json with Mermaid visualization

(Architecture diagram in README / web catalog.)

```
git clone <your-repo-url>
cd system-design-knowledge-graph
cp .env.example .env
# Edit .env locally only - never commit this file
uv sync --extra dev
```

2. Task & Link Manager - Monorepo

Assignment 1 | Full-stack | Complexity 50 | Info 45

<https://github.com/varunsood189/task-link-manager-extension>

Welcome to the Task & Link Manager, a powerful, developer-centric task management system designed to seamlessly bridge

Welcome to the Task & Link Manager, a powerful, developer-centric task management system designed to seamlessly bridge the gap between your browser and a persistent backend service.

Tech: Express, Chrome, Manifest V3, SQLite, Node.js, RAG

- Task & Link Manager Extension - A Manifest V3 Chrome extension providing an offline-first task management experience.
- Task & Link Manager Backend - An Express-based server with SQLite storage and a companion web portal.
- Frontend (Extension): Acts as the primary interface. It remains fully functional even when the server is offline, storing all data in the browser's local storage.
- Backend (Server): Acts as a centralized data vault. It allows you to backup your tasks and keep multiple browsers in sync. It also features its own web interface for managing tasks without the extension.

```
.
??? task-link-manager-backend/    # Node.js + Express server
?   ??? public/                  # Companion Web Portal
?   ??? db.js                    # SQLite database configuration
?   ??? server.js                 # API routes & sync logic
??? task-link-manager-extension/  # Chrome Extension (V3)
??? popup.js                     # Extension logic & sync engine
??? styles.css                   # Premium HSL-based design system
??? index.html                   # Extension popup interface
```

3. Stock News + Price Linking Agent (Chrome Side Panel)

Assignment 3 | Chrome Extension | Complexity 35 | Info 55

<https://github.com/varunsood189/stock-intelligence-platform>

This project is a Chrome Side Panel extension for multi-step stock research:
This project is a Chrome Side Panel extension for multi-step stock research:

Tech: Chrome, Gemini, Yahoo

- fetch latest company news,
- fetch stock prices for the same window,
- link news dates with price moves,
- return a 1-month buy/not-buy decision,

```
????????????????????????????????????????????????????????????
?                               Chrome Side Panel                      ?
?   UI + Agent Flow (popup_clean.js)                                ?
?   - Get Stock From Website                                         ?
?   - Run Multi-step Research                                         ?
?   - Test Direct Yahoo                                              ?
?   - Copy Trace                                                      ?
????????????????????????????????????????????????????????????
?                               ?
?                               ?
?                               ?
????????????????????????????????????????????????????????????
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
```

4. Assignment 4: MCP Server with Internet + CRUD + Prefab UI

Assignment 4 | MCP / AI Tools | Complexity 30 | Info 60

https://github.com/varunsood189/mcp_server_prefab

A working Model Context Protocol (MCP) server built with FastMCP, exposing three tools - internet fetch, local file CRUD, and a Prefab UI dashboard - to any MCP-compatible client (Claude, Cursor, or a custom script).

Tech: Gemini, MCP, FastMCP, Streamlit, Python, LLM, Prefab

Entry Points

```
??? fastmcp dev apps server.py    (browser, MCP protocol)
??? python client_demo.py         (stdio transport)
??? Any LLM MCP client            (MCP protocol)
|
v
server.py (FastMCP server)
??? fetch_company_ownership --> Wikipedia REST + Mediawiki APIs
??? local_file_crud         --> data/ folder (sandboxed)
??? ownership_dashboard     --> data/ folder (read)
--> Prefab browser preview
cd "assignment_mcp_prefab"
python -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
```

5. ResearchPulse

Assignment 2 | Chrome Extension | Complexity 15 | Info 45

<https://github.com/varunsood189/summerize-research-paper>

Chrome extension (Manifest V3) that summarizes research articles and research-related web pages, with optional **cit
Chrome extension (Manifest V3) that summarizes research articles and research-related web pages, with optional

citations and follow-up Q&A, using Google Gemini. The API key is never stored in this repository-only in your browser after you enter it in the extension settings.

Tech: Chrome, Manifest V3, Gemini, RAG

- No API keys, tokens, or secrets are hardcoded in the source files.
- Keys live only in chrome.storage.local on each user's machine after they save settings in the side panel.
- Before pushing, avoid committing browser export files, screenshots of settings, or any file that contains your real key.
- Google Chrome (or another Chromium browser with side panel support).

```
manifest.json      # Extension manifest
service-worker.js  # Gemini (see GEMINI_MODEL), shared generateContent helper, arXiv/PMC/citati
content-script.js  # Detects papers / page context on matched sites
sidepanel.html     # Side panel markup (summary type + citations strip)
sidepanel.js       # UI logic, storage, save/resume/history, citation fallbacks
sidepanel.css      # Styles
icons/             # Extension icons (add these locally)
```